

K. SLOAN LOUDEN-ROBINS

ELECTRICAL ENGINEER

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Portfolio:



LinkedIn:



EDUCATION (100% Self-Financed)

December 2025

Bachelor of Science in Applied Mathematics

Bachelor of Science in Electrical Engineering - Computer Engineering

Minor in Computer Science

Ingram School of Engineering – ABET Accredited Program – Texas State University – San Marcos, Texas

RELEVANT COURSEWORK

- Circuits 1 & 2
- Electronics 1 & 2
- Machine Learning
- VLSI Design
- C++ Coding 1 & 2
- Microprocessors

TECHNICAL SKILLS

- Simulation - LT Spice, Top Spice, Microwind, Multisim, MATLAB
- Languages - C++; Python; Unix/BASH
- Software – Microsoft Word, Excel, Teams, Google Drive, Vizio

PROJECTS

Project Manager

January 2025 – December 2025

Foundation Monitoring System, Ingram School of Engineering

- Documented the team's design and procedures, and delegated tasks to team members.
- Assigned weekly schedules with sponsors, mentee teams and civil teams.
- Analyzed costs, managed Bill of Materials (BOM), and balanced financial sheet.
- DRI for the data processing subsystem, interpolated and interpreted data, developed filtering algorithms implemented on central hub for varying sensors, researched vectoring scoring techniques and machine learning models. Streamlined file validation in Python and created S.O.P for hand off to next team.

Co-researcher

September 2025 - Present

Scatter Detector, TXST Space Lab

- Researched scatter detectors and nephelometers to design custom payload for a HAB system.
- Developed project timeline and created project proposal document.
- Designed: PCB in KiCad, scatter chamber in SolidWorks – coded payload for Co2 density measurements

EXPERIENCE

Undergraduate Research Assistant

January 2024 – May 2024

Mathematics Department Texas State University, San Marcos, TX

- Worked with Dr. Guillen to study sparse matrices and Fourier series decomposition

Lead Stylist

April 2013 – Present

LoLa Beauty, Austin, TX

- Received Glammy's award in 2017 and Best of the Knot Hall of Fame 2015-2024.
- Coordinated with assisting stylists, vendors and clients enabling efficient time allotment
- Resolved conflicts through effective communication and negotiation skills ensuring client satisfaction.

Learning Ambassador

June 2020 – June 2021

Amazon Warehouse (DAU2), Austin, TX

- Collected data and conducted surveys to propose ideas for overall workflow improvements
- Directed ~100 employees to ensure 60K+ packages are efficiently and safely prepared for delivery.

Assistant Manager and Stylist

April 2013 – May 2015

Gloss Beauty, Austin, TX

- Maintained inventory, managed multiple stylist's schedules, and balanced weekly finances
- Built and maintained clientele with a high retention rate.

ACTIVITIES

Member, Institute of Electrical and Electronics Engineers (IEEE)

September 2022 - Present

Member, Society of Women Engineers (SWE)

September 2024 – Present